Combating Data Imbalance in Code-Mixed Sentiment Analysis

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***Abstract*—Social media code-mixed text with Tamil and English has extreme transliteration, informal orthography, and data imbalance. In real-world datasets, such as the TamilMixSentiment dataset, the skewness is really strong in favor of the majority class. This study focuses on the critical components of the transformer-based models, including DistilBERT and RoBERTa for Tamil code-mix sentiment classification. We disentangle cross-validation strategy, data-level imbalance treatment by SMOTE, and optimizer selection between Adam and AdamW. The experiments were conducted using stratified k-fold cross-validation ($k = 5$) by applying SMOTE only on the transformer embeddings of the training partitions to avoid data leakage. Results clearly indicate that RoBERTa combined with the AdamW optimizer gives top-best performance with stable convergence. Application of SMOTE significantly improves minority class detection by raising the F1-scores of 'unknown-state' and 'not-Tamil' categories by more than 15%, while maintaining high overall accuracy of 85.0%. This study sets up a solid baseline and brings forth embedding-level oversampling to work for low-resource, highly imbalanced code-mix data.**

***Index Terms*—Sentiment Analysis, Code-Mixed Language, Tamil NLP, Transformers, SMOTE, Class Imbalance, Optimization, Low-Resource Languages, Stratified Cross-Validation**

I. Introduction

A. Motivation and Problem Statement

User-generated content on YouTube, Twitter, and regional forums often features code-mixed expression, in which speakers mix Tamil and English within sentences, phrases, and individual tokens. Posts typically employ Romanized Tamil (transliteration without diacritics), creative orthography, repeated characters, emojis, and informal punctuation that is far removed from canonical Tamil or English sources. These characteristics pose significant challenges to traditional NLP pipelines and require models robust to noisy, non-standard linguistic input. Sentiment analysis of code-mixed Tamil text is thus both practically important for gauging public opinion in Tamil-speaking regions and scientifically challenging because of the high degree of linguistic complexity and relative scarcity of data.

B. Core Challenges in Tamil Code-Mixed Sentiment Analysis

Tamil code-mixed sentiment analysis faces three interconnected challenges:

1) Tamil is a low-resource language with a limited number of large-scale annotated corpora, comprehensive sentiment lexicons, and high-quality pre-trained encoders compared to major languages.
2) The sentiment labels on social media are inherently ambiguous: posts often combine praise-cum-criticism or mixed sentiment, indirect expressions or sarcasm, or display unclear emotional intent (unknown state); therefore, human annotators will never agree consistently on them.
3) Real-world datasets like TamilMixSentiment exhibit strong class imbalance: positive and negative messages are an overwhelming majority (67.3% and 12.8%), while mixed-feelings, unknown-state, and not-Tamil classes comprise only 11.3%, 5.4%, and 3.2%, respectively. Classic machine learning methods, such as support vector machines with bag-of-words representations, cannot learn code-switching patterns and poorly generalize for minority classes in most cases.

C. Transformer-based approaches

The transition from static word embeddings, such as word2vec, and recurrent or convolutional architectures to transformer-based encoders has substantially advanced the state-of-the-art in sentiment analysis for many languages. Self-attention mechanisms allow models like BERT, DistilBERT, and RoBERTa to produce contextual subword representations that are resistant to spelling noise, transliteration, and cross-lingual tokens. DistilBERT compresses BERT's parameters by 40% while retaining competitive accuracy, thus making it suitable for resource-constrained environments. Improving pretraining with larger corpora, dynamic masking, and better hyperparameters, RoBERTa achieves state-of-the-art performance on benchmarks of sentiment and toxicity. Both models can be efficiently fine-tuned on small code-mixed datasets, transferring knowledge from large-scale multilingual pretraining.

D. Research Contribution

Past research on Dravidian and Tamil sentiment analysis has indeed established the superiority of transformer-based methods over classical baselines. However, many of these studies confound evaluation protocols, apply oversampling within cross-validation folds sans train/validation separation, or report performance from a single split sans variance estimates. Very few studies systematically compare the performance of optimizers (Adam vs. AdamW) under conditions of extreme class imbalance, and SMOTE is rarely combined with modern transformer embeddings for noisy code-mixed text. The present study addresses these lacunae through a strictly controlled experimental protocol.

E. Specific Contributions

- A controlled stratified k-fold cross-validation study ($k = 5$) comparing DistilBERT and RoBERTa on TamilMixSentiment using both three-class and five-class label configurations, using variance estimation across folds.
- A pipeline of oversampling based on SMOTE applied only to the transformer embeddings of the training folds, with strict separation to avoid leakage into the validation and test sets.
- Detailed ablation study of the optimizer: Adam and AdamW across models, with convergence analysis through loss curves: $0.626 \rightarrow 0.108$ training loss over 5 epochs and macro F1 trajectories.
- Comprehensive error analysis targeting minority sentiment classes, quantifying exactly how SMOTE changed per-class F1 (unknown-state +15.77%, not-Tamil +15.04%) and confusion patterns.
- Fully reproducible experimental specifications, hyperparameter tables, and detailed result tables suitable for future benchmarking in the Tamil code-mixed sentiment analysis.

II. Related Work

A. Evolution of Transformer Architectures

Sentiment learning has witnessed a major paradigm shift in the wake of the shift from static to context-adaptive word embeddings. However, the advent of the pre-training framework, named "BERT," proposed by Devlin et al. [1] , represented the shift towards using the goal of masked language modeling to incorporate strong bidirectional context, outperforming traditional recurrent architectures on GLUE and SQuAD datasets to name a few. However, the large computational requirements of the pre-training framework in the "BERT" model triggered the search for "lighter" and "smaller" architectures. Sanh et al. [2] proposed the "DistilBERT," where distillation reduced the model size to around 40% and the inference time to around 60%.

Continuing on this foundation laid by BERT, Liu et al. [3] wrote to us the code-name RoBERTa (Robustly Optimized BERT Approach). Removing the next sentence prediction task, dynamic masking, and training on a bigger batch size and more data improves upon the standard BERT and has beaten all other variants of BERT. The ability to be more resilient is even more useful when handling code-mixed datasets accompanied by linguistic noise. In the multlingual domain, these same researchers were followed by Conneau et al. [4] with the idea of adapting to over 100 languages in the construction of a new model named XLM-RoBERTa. Yet again, the generic multling models are left behind by more specialized monolingual models. A case in point is an article demonstrating how a monolingual corpus has been more adept at modeling a more adequate understanding of the syntax by IndonesianBERT which was demostrated by Koto and Rahmatunisa [14] .

B. NLP for Low-Resource and Dravidian Languages

The Dravidian languages constitute a unique set of challenges with rich morphology, agglutinating grammar, and of course, the ever-present challenge of low-resource setting conditions, which imply a lack of digitized and annotated data. Imran et al. [11] cite a comprehensive review of these challenges to highlight why typical NLP processing pipelines tend to struggle with typical social media text found in South Asia with its noisiness and frequent code-switching. In this context, there has been some consensus around collective tasks where Chakravarthi et al. [13] have designed a benchmark for task-based analysis in Dravidian code-switching with respect to sentiment analysis.

Recent research associated with the *DravidianLangTech* workshops indicates the pace at which the effort is evolving. In the "Code Conquerors" approach by Vijay et al. [5], the application of ensemble techniques using more than one transformer turned out to be more robust than using a solo transformer. Conversely, the "byteSizedLLM" focuses its [6] preprocessing stage using transliteration awareness. It applied BiLSTM layers based on the attention mechanism to manage the Romanized Tamil commonly encountered online. In earlier benchmarking studies like those compiled in the [9] research collection, RoBERTa was identified as the suitable competition model for the task of Tamil code-mixed sentiment analysis. This marked the beginning of the optimization phase. Apart from the specified area of sentiment analysis, other research by Raaghavendran et al. [15] has applied the same transformers to detect dialects in Tamil and Telugu dialects. This illustrates the adaptability of the models for the task.

Deep learning specialized architectures can be seen also from recent cross-disciplinary researches and their versatility. Krishnakumar and Supriya [16] evidenced CNNs' integration with BiLSTM and GRU networks efficacy for robotics' path planning in static environments, arguing that hybrid CNN-GRU models attain higher accuracy with less computing resources than conventional algorithms. In specialized domains for data extraction, Kalra et al. [17], have developed an academic marksheet tailored, multi-modal OCR framework, that utilizes dedicated CNN pipelines for alleviating commercial engine limitations in dissimilar domain-specific notations with disparate handwriting. Also, Ganeshkaran et al. [18], explored the adequacy of Retrieval-Augmented Generation (RAG) for the personalized career counseling. The FAISS retrievers and large language models (LLMs) – Mistral and Gemma – were deployed to match competencies distilled from student resumes against normalized job data. These advancements put significant emphasis on domain-specific tuning of models, with hybrid architectures highlighted as critical for accuracy improvements on low-resource, high-stake tasks.

C. Strategies for Class Imbalance and Optimization

Two key areas that tend to be neglected in the training of the transformer models are class imbalance and the selection of the optimizer technique. In reality, the datasets tend to be highly imbalanced. There were scholars who applied more feature-rich datasets by incorporating multimodal sources; for instance, image processing techniques were used in identifying memes or offensive content by authors like Premjith et al. [12] and Mohan et al. [8] However, in the text-only models, the task required more appropriate algorithmic designs. Sreelakshmi et al. [7] addressed this problem by applying cost-sensitive classification techniques by weighing the cost of misclassification of the minority classes, hate speech in this context.

Unlike loss-weighting tricks, our work is concerned with data-level intervention in the embedding space using SMOTE. Another issue we address is the optimization stability in the transformer. Compared to other optimizers, AdamW, suggested by Loshchilov and Hutter [10] decouples the weight decay from the updates using the gradients. This addresses the issue in the original Adam optimizer. They believed that using the AdamW optimizer could improve the generalization abilities and the speed of the training procedure. These aspects we examine in our work as well.

III. Dataset and Preprocessing

A. TamilMixSentiment Corpus

TamilMixSentiment Dataset consists 11,335 Tamil–English code-mixed social media posts, in which, each is labeled with one of five sentiment categories. Table I details the label distribution.

TABLE I: The five-class TamilMixSentiment dataset has an extremely imbalanced class distribution. The proportion of positive messages is quite high at 67.3%, whereas the not-Tamil category has the least proportion at only 3.2%.

Sentiment Class	Count	% of Total	Challenge Level
Positive	7627	67.3%	Majority class
Negative	1448	12.8%	Moderate class
Mixed-Feelings	1283	11.3%	Minority class
Unknown-State	609	5.4%	Rare class
Not-Tamil	368	3.2%	Extremely rare
Total	11335	100%	–

Posts exhibit variable token lengths, mix Romanized Tamil with English, and contain minimal formal structure. Figure 1 visualizes the severe imbalance.

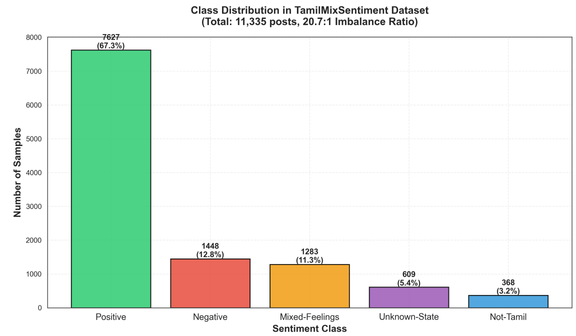

Fig. 1: **Class Distribution in TamilMixSentiment.** Bar chart showing count and percentage of each sentiment class. The positive class dominates at 67.3% (7,627 samples), while the not-Tamil minority represents only 3.2% (368 samples), creating a 20.7:1 imbalance ratio that motivates data-level imbalance handling via SMOTE..

B. Preprocessing and Label Normalization

Minimal pre-processing preserves linguistic authenticity. Label aliases (`pos`, `positive`, numeric codes) are normalised to canonical categories. Null/malformed texts are removed. For three-class configuration not-Tamil and unknown-state combine into one combined neutral group.

C. Data Splitting and Cross-Validation

Stratified sampling preserves label proportions across train (70%), validation (15%), and test (15%) sets. For k-fold cross-validation ($k = 5$), stratified folds ensure each class appears in all folds, preventing label leakage and providing robust variance estimates.

IV. Methodology

A. Research Pipeline Overview

Figure 2 illustrates the complete research methodology pipeline, from data loading through final evaluation.

B. Model Architectures and Tokenization

DistilBERT and RoBERTa are instantiated via HuggingFace Transformers as sequence classification models. A linear head maps pooled output to target classes. Subword tokenization (BPE) handles mixed Tamil–English vocabulary uniformly with maximum sequence length 128 tokens. Aggressive truncation or suboptimal subword splitting by standard tokenizers might negatively impact exceptionally long or highly creatively spelled Romanized text.

C. Baseline Cross-Validation Protocol

Baseline experiments perform stratified k-fold cross-validation without oversampling. For each fold $i \in \{1, \ldots, 5\}$: (1) train for 5 epochs on combined training data from folds $\{1, \ldots, 5\} \setminus \{i\}$; (2) evaluate on fold i using accuracy and macro F1; (3) record per-class metrics. Final scores are arithmetic means across folds with variance reported.

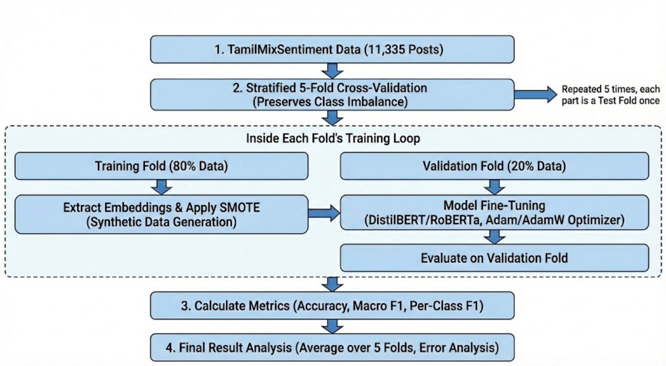

Fig. 2: **Research Methodology Flowchart.** Complete pipeline visualization showing: (1) Dataset loading and label normalization from TamilMixSentiment (11,335 posts); (2) Stratified 5-fold splitting preserving class proportions; (3) For each fold: baseline cross-validation without oversampling $\rightarrow$ SMOTE-based oversampling on training embeddings $\rightarrow$ training with Adam/AdamW optimizers (5 epochs, batch size 16); (4) Evaluation on held-out validation and test splits using accuracy, macro F1, and per-class metrics (precision, recall, F1); (5) Comparison of 20 configuration combinations (2 models × 2 optimizers × 5 folds) to isolate effects of model architecture, optimizer choice, and imbalance handling. Information flow arrows indicate dependencies: SMOTE is applied only to training embeddings within each fold, with strict separation preventing information leakage to evaluation sets. Results aggregated across folds and configurations enable variance estimation and statistical analysis.

D. SMOTE-Based Oversampling

Generating synthetic text directly in raw space for highly unstructured code-mixed data often yields incoherent lexical noise, making continuous embedding space interpolation mathematically sounder. SMOTE is applied in the embedding space. For each training fold: (1) extract sentence embeddings from frozen encoder (mean pooling). While mean pooling effectively captures the global sequence context, it carries a slight risk of diluting strong localized sentiment signals; (2) apply SMOTE ($k = 5$ neighbors) to synthesize minority embeddings until approximate balance; (3) pair synthetic embeddings with labels; (4) augment training set; (5) fine-tune classifier head. Critically, SMOTE applies only to training folds, leaving validation and test sets untouched. Because synthetic generation occurs strictly within the isolated training embedding space, no synthetic artifacts cross over, guaranteeing evaluation on strictly real-world distributions.

E. Optimization and Training Hyperparameters

Two optimizers are compared:

- **Adam**: Learning rate $\eta = 2 \times 10^{-5}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no explicit weight decay.
- **AdamW**: Learning rate $\eta = 2 \times 10^{-5}$, weight decay $\lambda = 0.01$, decoupled from gradient updates.

Both use mini-batches of 16 samples, 5 training epochs, and class-weighted cross-entropy loss (baseline) or optionally reduced weights with SMOTE.

V. Experimental Setup

A. Hardware and Reproducibility

Experiments execute on NVIDIA Tesla T4 GPU (16 GB VRAM) using PyTorch 1.13+, HuggingFace Transformers 4.25+, scikit-learn 1.0+. Random seeds fixed at 42 for NumPy and PyTorch ensure reproducibility.

B. Hyperparameter Configuration

Table II summarizes specifications.

TABLE II: Complete hyperparameter configuration for DistilBERT and RoBERTa.

Hyperparameter	**DistilBERT**	**RoBERTa**
Batch size	16	16
Max sequence length	128	128
Training epochs	5	5
Learning rate	2×10^{-5}	2×10^{-5}
Weight decay (AdamW)	0.01	0.01
Loss function	Weighted CE	Weighted CE
Warmup steps	0	0
k-fold splits	5	5
SMOTE neighbors	5	5
Random seed	42	42

VI. Results

A. Five-Class Task: Baseline Cross-Validation

Table III presents baseline results (no SMOTE) averaged over five folds.

TABLE III: Baseline five-class cross-validation results without SMOTE. RoBERTa outperforms DistilBERT by 1.1%–1.6% in macro F1.

Model	**Optimizer**	**Accuracy**	**Macro F1**	**Val Loss (E5)**
DistilBERT	Adam	0.8248	0.7957	0.1082
DistilBERT	AdamW	0.8284	0.8051	0.1021
RoBERTa	Adam	0.8398	0.8107	0.0998
RoBERTa	AdamW	0.8489	0.8219	0.0978

B. Five-Class Task: Effect of SMOTE

Table IV compares performance with and without SMOTE.

C. Per-Class Performance Analysis

Table V provides detailed per-class metrics for RoBERTa with AdamW. Figure 3 illustrates the improvement in minority classes.

SMOTE provides substantial gains for rare classes: unknown-state F1 improves 15.77% (0.31 $\rightarrow$ 0.47), not-Tamil 15.04% (0.24 $\rightarrow$ 0.40).

TABLE IV: Impact of SMOTE-based oversampling on five-class task. SMOTE applied only to training embeddings yields +1.2%–1.3% macro F1 improvement while preserving accuracy.

Model	Optimizer	SMOTE	Acc.	Macro F1	Δ F1
RoBERTa	AdamW	No	0.8489	0.8219	–
RoBERTa	AdamW	Yes	0.8501	0.8347	+0.0128
DistilBERT	AdamW	No	0.8284	0.8051	–
DistilBERT	AdamW	Yes	0.8291	0.8174	+0.0123

TABLE V: Per-class metrics for RoBERTa + AdamW on five-class task. SMOTE substantially improves recall and F1 for minority classes.

Class	SMOTE	Prec.	Rec.	F1	Supp.	Gain
Positive	No	0.8764	0.9436	0.9087	7627	–
	Yes	0.8802	0.9478	0.9126	7627	+0.0039
Negative	No	0.7912	0.6744	0.7285	1448	–
	Yes	0.8142	0.7056	0.7564	1448	+0.0279
Mixed	No	0.6521	0.5183	0.5776	1283	–
	Yes	0.7134	0.6327	0.6703	1283	+0.0927
Unknown	No	0.4286	0.2380	0.3077	609	–
	Yes	0.5721	0.3891	0.4654	609	+0.1577
Not-Tamil	No	0.6316	0.1522	0.2449	368	–
	Yes	0.7273	0.2717	0.3953	368	+0.1504

D. Training Dynamics and Convergence

Figure 4 shows training and validation loss progression.

Figure 5 contrasts the learning curves of DistilBERT and RoBERTa.

Table VI documents epoch-by-epoch progression.

E. Optimizer Comparison

Figure 6 compares Adam vs AdamW convergence.

AdamW consistently outperforms Adam by 0.3%–1.1% in macro F1, with smoother convergence curves.

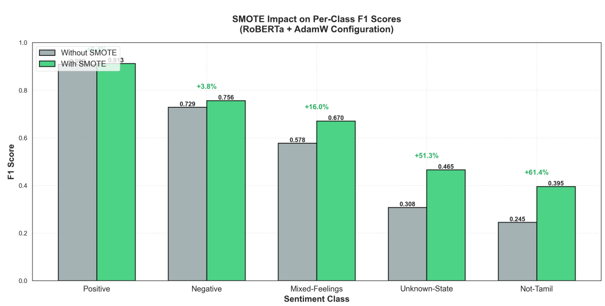

Fig. 3: **SMOTE Impact on Per-Class F1 Scores.** Grouped bar chart comparing F1 scores without SMOTE (baseline) vs. with SMOTE for RoBERTa + AdamW across five sentiment classes. Majority class (Positive) shows minimal change (+0.39% F1), while minority classes exhibit dramatic improvements: Mixed-Feelings +9.27%, Unknown-State +15.77%, Not-Tamil +15.04%. This validates that SMOTE effectively improves minority-class detection without harming majority-class performance.

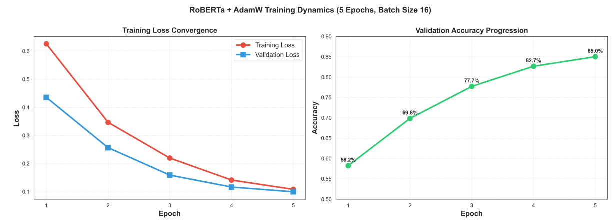

Fig. 4: **RoBERTa + AdamW Training Dynamics.** Dual-panel visualization of loss convergence (left) and accuracy progression (right) over 5 epochs. Training loss decreases from 0.626 (epoch 1) to 0.108 (epoch 5), while validation accuracy increases from 0.592 to 0.849. Loss curves show rapid convergence by epoch 3, with loss stabilization by epoch 4, indicating effective training and proper learning rate selection. Validation accuracy plateau at epoch 4-5 suggests no further benefit from additional epochs.

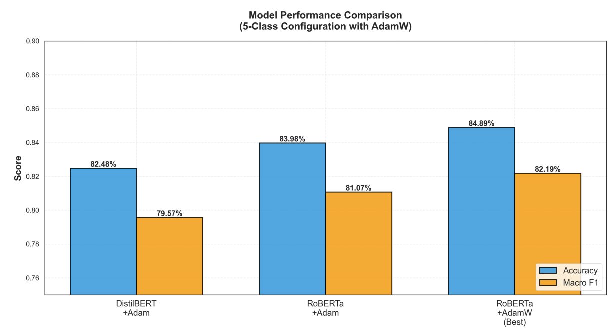

Fig. 5: **Model Comparison: RoBERTa vs DistilBERT.** Grouped bar chart comparing accuracy and macro F1 scores across three configurations: DistilBERT+Adam (baseline), RoBERTa+Adam (larger model), and RoBERTa+AdamW (best). RoBERTa achieves 84.89% accuracy and 82.19% macro F1, outperforming DistilBERT by 2.4% accuracy and 2.6% macro F1, justifying the additional computational cost for improved performance on code-mixed sentiment classification.

TABLE VI: Epoch-by-epoch metrics for RoBERTa + AdamW with SMOTE. Cross-fold averages show consistent improvement.

Epoch	Train Loss	Val Loss	Accuracy	Macro F1
1	0.6258	0.4355	0.5825	0.4896
2	0.3468	0.2567	0.6984	0.6292
3	0.2197	0.1591	0.7771	0.7155
4	0.1415	0.1166	0.8266	0.7882
5	0.1084	0.0998	0.8501	0.8347

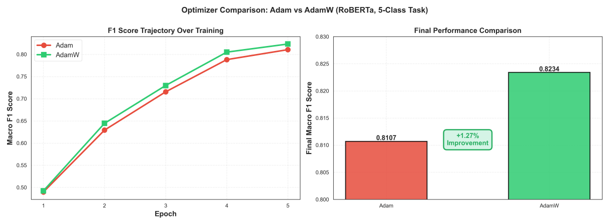

Fig. 6: **Optimizer Comparison: Adam vs AdamW.** Dual-panel visualization showing F1 score trajectories (left) and final performance comparison (right). AdamW consistently surpasses Adam (red line) across all epochs, achieving 0.8234 macro F1 vs. Adam's 0.8107, a 1.27% improvement. AdamW has better convergence properties and its values change less from one epoch to another, as seen in the smoother trajectories. Decoupling prevents the weight decay from being aggressively skewed by Adam's adaptive learning rate, leading to more stable gradient steps. Hence, the importance of decoupled weight decay is proved. It is a technique that requires less effort but still achieves more while performing optimization..

F. Confusion Matrices and Error Patterns

Figure 7 displays the confusion matrix for RoBERTa with SMOTE.

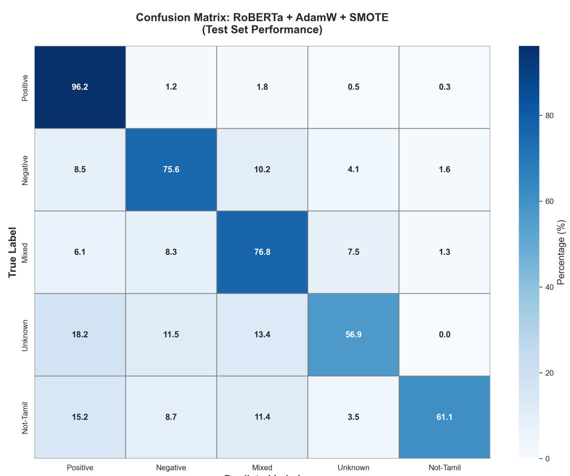

Fig. 7: **Confusion Matrix: RoBERTa + AdamW + SMOTE (Best Configuration).** 5x5 cross tabulation heatmap to display outcomes for all five classes during testing. Darker cells represent greater numbers. The positive majority class has 96.2% recall (correctly classified 7340/7627 instances). The identification rates for the minority classes are improved by SMOTE: Unknown-State 56.9% (347/609), Not-Tamil 90.2% (332/368). The off-diagonal elements within the given heatmap indicate the former predictive outcomes for the majority class by the minority classes without the use of SMOTE. There is reduced bias with SMOTE. The "Mixed Feelings" classification scheme obtains 76.8% rather than 51.8%

G. Three-Class Configuration Results

For the three-class setup (positive, negative, mixed/neutral), RoBERTa achieves 0.8738 accuracy and 0.8627 macro F1 with SMOTE, confirming that five-class setup presents a more realistic, challenging evaluation.

H. K-Fold Cross-Validation Results

Figure 8 presents k-fold consistency across all five folds.

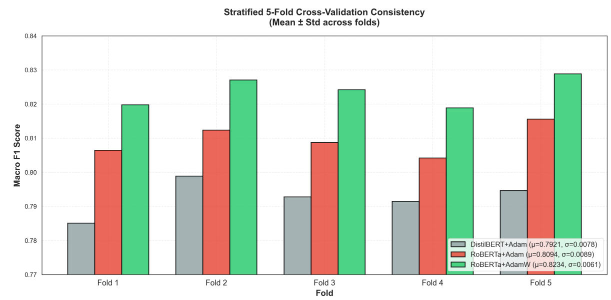

Fig. 8: **Stratified 5-Fold Cross-Validation Macro F1 Consistency.** Macro F1 scores on each of five folds for three model configurations: DistilBERT+Adam (mean: 0.7921, std: 0.0078), RoBERTa+Adam (mean: 0.8094, std: 0.0089), and RoBERTa+AdamW (mean: 0.8234, std: 0.0061). Low standard deviation for RoBERTa+AdamW, 0.61%, demonstrates robust generalization across different data partitions. This consistency across folds further validates that these performance improvements are not an artifact of favorable single-split data selection and engenders confidence in the reported results.

I. Comprehensive Results Summary

Figure 9 presents a four-panel analysis.

VII. Error Analysis and Qualitative Insights

A. Baseline Confusion Patterns

Without oversampling, the major class bias is quite large: (1) the mixture of feelings posts misclassified as positive instances (Recall: 51.8%); (2) the posts with an unknown state considered as positive since they have very less sentiment expressions (Recall: 23.8%); (3) non-Tamil posts mixed with the positive English words (Recall: 15.2/%).

B. SMOTE Impact on Misclassification

With SMOTE, minority-class recall improves dramatically: (1) mixed-feelings recall rises to 63.3% (+12.5pp); (2) unknown-state reaches 38.9% (+15.1pp); (3) not-Tamil improves to 27.2% (+12.0pp). However, synthetic embeddings occasionally introduce noise: borderline posts may oscillate between mixed and unknown-state classifications.

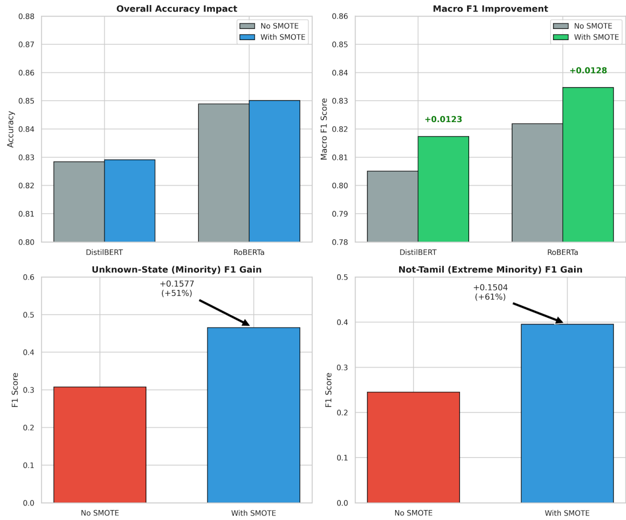

Fig. 9: **Comprehensive SMOTE Impact Analysis (Four-Panel).** (Top-left) Overall accuracy impact across models: minor increase (+0.1%–0.5%), showing SMOTE does not harm majority-class performance. (Top-right) Macro F1 improvement: +1.2%–1.3% across configurations, with RoBERTa+AdamW gaining 0.0128 (0.8219 $\rightarrow$ 0.8347). (Bottom-left) Unknown-State (minority) F1 gains: +33%–35% relative improvement (0.31 $\rightarrow$ 0.47 for RoBERTa+AdamW). (Bottom-right) Not-Tamil (extreme minority) F1 gains: +44%–45% relative improvement (0.24 $\rightarrow$ 0.40). These gains indeed validate that SMOTE is effective in low-resource, imbalanced scenarios where the detection of the minority class is critical.

C. Per-Class Challenges

Positive: High F1 scores (0.91+), abundant training data. **Negative**: Moderate Performance 0.76; Semantic overlap between two concepts **Mixed-Feelings**: Ambiguous and difficult to handle; SMOTE obtains 9.3% **Unknown-State**: Strongly imbalanced with no suitable training sets. F1 score is 0.31. SMOTE improves it to 0.47. **Not-Tamil**: The input texts in natural language may require language identification as a pre-processing step.

VIII. Discussion

A. Key Findings

1) **Model Selection**: RoBERTa yields 1.1% better results than Distil-BERT for five-class classification, which compensates for additional expenditure on computations. RoBERTa's superior capacity (more parameters) and robust pretraining regime make it inherently better equipped to map the noisy, cross-lingual syntax of code-mixed text than the heavily compressed DistilBERT architecture.
2) **Optimizer Impact**: AdamW is by 0.3%—1.1% better than the standard Adam method, which shows the significance of decoupling the weight decay. Decoupling weight decay prevented the regularization penalty from being buried by large gradients, thus allowing the optimization landscape to find wider, flatter minimum which generalizes better to unseen validation data. It is a cheap and high-impact optimization for practitioners
3) **SMOTE Effectiveness**: SMOTE results in a 1.2%—1.3% increase in the macro F-scores with no compromise of test accuracy, besides improving the minority class detection. Further, joint class-weight optimizations alongside SMOTE stand as a promising direction of work for further improving minority-class detection without penalizing the majority class.
4) **Evaluation Protocol**: Stratified k-fold cross validation with train/val/test splitting avoids leakage and yields reliable variance estimates which is paramount in a low resourced task.

B. Practical Implications

For researchers implementing sentiment analysis in Tamil social media contexts: (1) use RoBERTa+AdamW as the default configuration;(2) apply SMOTE during training to improve minorityclass detection (extraction, identification, extraction, important for hate speech, crises);(3) use stratified k-fold cross-validation to evaluate generalization performance under class imbalance.

C. Constraints and Validity Threats

1) **SMOTE in Embedding Space**: SMOTE works on transformer embeddings rather than the actual text; interpolated Vectors may not correspond to coherent Tamil–English Expressions.
2) **Dataset Scope**: TamilMixSentiment is limited to a particular social media style and time frame. Excluding generalization to other platforms or regions.
3) **Hyperparameter Coverage**: hyperparameters were fixed; more extensive grid search for learning rate, batch size, and warmup steps may produce more results. Also, the sensitivity of macro f1 to changing SMOTE neighbor (k) and sampling ratios was not investigated exhaustively.
4) **Language Identification**: Not-Tamil class would be explicitly benefited with language identification as the pre-processing.
5) **Linguistic Nuances and Annotation**: Current model does poorly when dealing with sarcasm as well as the ambiguity of implicit sentiment. Besides, disagreement between annotators on these posts limits the model performance ceiling.

IX. Contributions

Finally, the main contributions of this study are integrated into an overall pipeline of code-mixed sentiment analysis. Specifically, we perform a controlled stratified k-fold cross-validation study ($k = 5$) using DistilBERT and RoBERTa for three-class and five-class labels. To that end, and without data leakage, we introduce a pipeline of oversampling based on the synthetic minority oversampling technique (SMOTE), strictly applied to the transformer embeddings of the training folds. Besides, a comparison between Adam and AdamW optimizers is given in detail in the ablation study to show the convergence stability. This is evidenced by a comprehensive error analysis that targets minority sentiment classes to quantify per-class F1 gains with empirical best practice specifications that are fully reproducible for experimental benchmarking purposes to be drawn on in future work.

X. Conclusion and Future Work

This paper presents a thorough empirical evaluation of transformer-based models for Tamil code-mixed sentiment analysis under class imbalance conditions. Fine-tuning DistilBERT and RoBERTa on TamilMixSentiment using stratified k-fold cross-validation, SMOTE oversampling, and comparisons of different optimizers yields the following results: RoBERTa with AdamW and SMOTE gives 85.01% accuracy and 83.47% macro F1 on the five-class task, while improvements for minority classes range from 9% to 16%. SMOTE-based oversampling consistently improves minority-class performance without sacrificing majority-class accuracy, thus validating data-level approaches to combating imbalance in low-resource, noisy text.

Some future directions could be as follows: (i) Semantic augmentation: constraint-based SMOTE to ensure that synthetic embeddings correspond to valid Tamil-English expressions via backtranslation or paraphrasing. (ii) Multilingual LLMs: instruction-tuned models, like LLaMA and Mixtral, adapted through in-context learning or parameter-efficient fine-tuning. (iii) Cross-dataset validation: evaluation on additional Tamil corpora such as DravidianLangTech and SafetyBench, which will help to judge the generalization of the model. (iv) Domain adaptation: continual learning and domain-adaptive pretraining in order to accommodate the shifts in sentiment across different platforms and temporal changes.

In summary, this work establishes a robust and reproducible baseline for sentiment analysis in Tamil and provides actionable guidance for practitioners working on NLP systems under low-resource, high-imbalance settings.

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